



The Spaghetti Engine
VISUAL SYSTEM MODELLING WORKSPACE

The Spaghetti Engine

Development Log

Fuzzy Cognitive Map modelling tool

Field	Value
Version	0.4.1 (prototype)
Log entry date	8 October 2026
Status	Built and verified locally; not yet deployed. Version 0.4.0 remains the live release until this one is pushed.
Live URL	spaghetti-engine.netlify.app (Netlify, auto-deployed from the GitHub main branch)
Source repository	github.com/aPaarl/Spaghetti-engine (release commit to be recorded when 0.4.1 is pushed to main)
Previous version	0.4.0, 26 September 2026, commit b11f3dc (see Spaghetti_Engine_Dev_Log_v0.4.0.docx)
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1. Purpose of This Log

This Development Log records the changes from version 0.4.0 (26 September 2026) to version 0.4.1 of The Spaghetti Engine. Version 0.4.1 is a small release: it changes one convention, the direction of a decrease, on the Transition Point Analysis and Transition Pathways tabs, so that a decrease in an analysis is the same intervention as a decrease in a scenario.

The log documents what changed and why, what it does to results obtained with the earlier convention, how the change was verified, and the deployment status. It is a technical companion to the User Manual and the Methodological Annex, both updated to version 0.4.1 alongside this log (Section 10).

2. Summary of Version 0.4.1

Area	Change
Transition Point direction	A decrease now sweeps the intervention from 0 down to -1 instead of from 1 down to 0. An increase is unchanged (0 up to +1). The intensity axis still runs from 0 to 1: an intensity x holds the intervention at x in an increase and at $-x$ in a decrease.
Transition Pathways direction	A decreasing lever follows the same rule: intensity u holds it at $-u$. Before, intensity u held it at $1 - u$, so intensity 0 meant a lever already at +1 and full intensity a lever at 0.
Consistency with scenarios	The last step of each sweep is exactly the scenario in which the concept is a driver at +1 (increase) or at -1 (decrease), and the first step is the concept held at 0 in both directions.
Labels and exports	Buttons, headings, explanations, the Intensity tooltip, and the Direction columns of the exports say Decrease (0 to -1).
Documentation and repository	Manual, annex, and this log are issued as 0.4.1. A regression test for the change (tests/test_transition_direction.cjs) and a ROADMAP.md are added to the repository.

3. The Change in Detail

3.1 Transition Point Analysis

- **Why.** A decrease from 1 to 0 was the increase curve read backwards (on the peat meadow example the two differ by exactly 0), so it added no information. It also never tested the state a scenario creates when it drives a concept down to -1, the other way a scenario can drive a concept.
- **What it does now.** `runTransitionSweep` locks the intervention at x for an increase and at $-x$ for a decrease, for x from 0 to 1 in equal steps, and reads the equilibrium of every chosen outcome at each step. The effect is the change from the first step (the intervention held at 0) to the last (held at +1 or -1).
- **Reading the results.** Intensity, transition point, response type (equal thirds), efficiency, and the rankings keep their meaning, with intensity as the distance from 0. For a decrease the slope is taken with respect to intensity,

so a positive slope means that the outcome rises as the intervention is pushed further down.

- **Reference state.** As before, effects are measured from the intervention held at 0, not from the baseline in which it is free to move. The Method & diagnostics panel now says so.

3.2 Transition Pathways

- **What it does now.** A lever's activation comes from one helper, pathwayLeverValue: u for an increase and $0 - u$ for a decrease. The pathway engine and the order-invariance test both use it. Intensity 0 is therefore the neutral lever in both directions, and full intensity is the lever held at +1 or -1.
- **Before.** A decrease held the lever at $1 - u$, so its profile began with a full push (the lever at +1 at intensity 0) and ended at 0, and a stage that implements a lever up to its transition point had no neutral starting point.
- **Unchanged.** The gain measure (total outcome gain, TOG), the transition-point rule (the largest positive marginal return in TOG, reported at the upper end of the step), the effort weights, the archetypes, and everything for increasing levers.

3.3 Labels and exports

- **Interface.** The Step 2 buttons read Increase (0→1) and Decrease (0→-1); the explanation box, the chart headings and exported figure titles, and the Method & diagnostics note follow the new convention. On the Transition Pathways tab the lever buttons and the Intensity tooltip say the same.
- **Exports.** The Direction column of the Transition Point CSV now reads Increase (0 to 1) or Decrease (0 to -1); before, it said only increase or decrease. The pathway lever tables and workbook sheets say Decrease (0 to -1).

4. Implementation Choices

Where the change needed a decision, the choice and its reason are recorded here.

Topic	Choice in the tool	Reason
Intensity axis	Stays 0 to 1 in both directions; a decrease holds the concept at $-x$.	Classification, efficiency, and rankings keep their meaning, and the two directions can be compared on one scale.
A withdrawal option	Not offered. Going from 1 down to 0 is the increase curve read backwards.	It would show nothing new; the question of what happens when full effort is withdrawn is answered by reading the increase curve from the right.
Signed zero	A decrease holds the concept at $0 - x$, not $-x$.	So that the first step is $+0$ and not -0 in exports and comparisons.
One helper for lever values	pathwayLeverValue is used by the pathway engine and by the invariance test.	The two cannot drift apart.
Saved pathway setups	Not migrated.	A setup saved in the browser keeps its decrease levers, which now mean 0 to -1. Results are re-run on demand, so nothing stored goes stale.

Topic	Choice in the tool	Reason
Release number	0.4.1, not a later 0.4.0 build.	A change of convention that alters results for decreasing levers is reported as a release, even though no feature is added.

5. Effect on Existing Results and Saved Setups

- Transition Point Analysis, increase: no change.
- Transition Point Analysis, decrease: new results. The earlier decrease results are the increase results read backwards, so nothing is lost; run a decrease again to see how the outcomes respond when the intervention is pushed down.
- Transition Pathways with a decreasing lever: results change (lever profile, transition point, gain at full intensity, the pathways built from it, and the invariance test). A paper, report, or slide that used a decreasing lever should be run again. Analyses with only increasing levers are unaffected.
- Saved setups: kept as they are, with decrease now meaning 0 to -1.
- Documentation: the 0.4.0 manual, annex, and log describe the previous convention.
- User Manual: the typographic minus sign was replaced by a plain hyphen throughout, as in earlier hand edits, so that the -1..+1 scale reads the same everywhere.
- Version raised from 0.4.0 to 0.4.1 in package.json and package-lock.json. No other analytical method or setting changed.

6. Defects Found and Fixed

Problem	Cause	Fix
A decrease on the Transition Point tab repeated the increase curve and never reached the state of a scenario that drives the concept down.	The sweep ran the intervention from 1 to 0, locking it at 1 - x.	The sweep runs from 0 to -1, locking the concept at -x; its last step is the scenario with the concept held at -1.
A decreasing lever on the Transition Pathways tab was already fully pushed at intensity 0.	Its activation was 1 - u, so intensity 0 held it at +1.	Activation is -u; intensity 0 is the neutral lever, as for an increasing lever.
The tooltip on the Transition Pathways tab said a decrease worked as on the Transition Point tab, which the first change would have made untrue.	Wording written for the earlier convention.	Reworded to state the new convention for both tabs.

Problem	Cause	Fix
The Direction column of the Transition Point CSV did not say what range the intervention covered.	It stored the setting (increase or decrease), not the range.	It states the range, for example Decrease (0 to -1).

7. Verification

- **Build:** the production build (vite build) completes without errors.
- **New test (tests/test_transition_direction.cjs, 25 checks):** it loads the engine straight from the source and runs on the water-energy-food and peat meadow examples. The last step of each Transition Point sweep equals the scenario in which the concept is a driver at +1 or at -1, for every outcome (difference below 10^{-12}); both directions start from the concept held at 0; the effect of a decrease equals the change from 0 to the scenario at -1; and a decrease is no longer the increase curve read backwards. On the Transition Pathways engine, the lever profile has the same number of points in both directions, full intensity equals the lever locked at +1 or -1, intensity 0 equals the lever locked at 0, and every point of a decrease profile equals the lever locked at -u. All 25 pass.
- **Earlier suites:** the engine tests of 0.4.0 (119 tests, with the randomized engine check and the static checks) were not rerun, because the scripts were kept in temporary folders that were cleaned up afterwards. The change touches only the sweep and the lever-activation mapping, not the engine those suites cover. Only the new test is in the repository; rebuilding the full set there is on the list of next steps.
- **The two directions differ:** in the peat meadow example (for the lever Raised groundwater levels), 11 of 19 outcomes respond differently to a push down than to a push up, meaning that the effect of the push down is not the sign-reversed effect of the push up, to within 0.05; in the water-energy-food example, 24 of 39. For example, pushing Raised groundwater levels up from 0 to +1 moves Drainage depth by -1.77 (from 0.80 to -0.97), whereas pushing it down to -1 moves it by only +0.16 (to 0.96): a one-sided response that the earlier decrease showed only as a mirror image.
- **In the running app:** a Transition Point analysis with a decrease on the water-energy-food example (button, headings, results table, explanation text, and the Method & diagnostics note), and a Transition Pathways run with a decreasing lever (lever profiles, rankings, invariance test), completed without errors in the console other than the known deprecation warnings of the chart library. The CSV and workbook columns were checked in the source, not downloaded.

8. Deployment Status

- **Not yet deployed.** The 0.4.1 changes exist only in the local working copy. Nothing has been pushed; 0.4.0 (commit b11f3dc) remains live. When the release is pushed to main, Netlify rebuilds automatically, and the commit and the live bundle should be recorded here.
- **Files changed:** src/SpaghettiEngine.jsx (the two sweeps, labels, explanations, exports), package.json and package-lock.json (version), README.md, ROADMAP.md, tests/test_transition_direction.cjs, tools/docx_edit.py (a helper for editing the Word documents), src/docs-manifest.json, the three PDFs in public/docs, and public/sitemap.xml.

- **PDF files:** the three 0.4.0 PDFs in public/docs are replaced by the 0.4.1 versions, and the sitemap lists the new file names. The 0.4.0 documents remain in the version folder on the share and in the repository history.
- **Build note:** npm run build fails in the local OneDrive folder because npm's command shim cannot handle the ampersand in the folder path; running node node_modules/vite/bin/vite.js build directly works.

9. Prototype Scope and Known Limitations

- All limitations listed in Section 9 of the 0.4.0 log still apply.
- The Transition Point tab no longer has a direction that starts from full effort (1 to 0). That reading is available by reading the increase curve from the right, with the transition point at one minus the reported value.
- Saved pathway setups are not migrated: a lever saved as a decrease now means 0 to -1.
- The regression tests of 0.4.0 are not in the repository (see Section 7).

10. Related Documentation

Document	Covers
The Spaghetti Engine - User Manual (version 0.4.1)	How to use every feature. Section 1.2 lists the changes of 0.4.1; Sections 12 and 13 describe the new direction.
The Spaghetti Engine - Methodological Annex (version 0.4.1)	The methods. Sections 12.1 and 13.1 define the sweep and the lever activation under the new convention.
Transition Pathway Analysis - methods extension (draft)	The methodological framework the Transition Pathways tab is based on; it does not describe the earlier direction convention.
Spaghetti_Engine_Dev_Log_v0.4.0	The 0.4.0 release: the Monte Carlo tab, tab groups, the new branding, and the About & documentation tab.
This Development Log	What changed in 0.4.1 and the technical work behind it.

11. Next Steps

- Rebuild the full engine regression set (the 0.4.0 suites and the randomized engine check) inside the repository, in tests/, so that it survives between sessions.
- Run again any pathway results that used a decreasing lever, for example in the EMS article draft.
- Continue with the development roadmap: version 0.5.0 (guided first use) is next, then easier modelling, clearer analysis, performance, pathways toward an envisioned future, and version 1.0.
- Bring the methods extension in line with the choices of the 0.3.0 log (iteration cap, sweep-resolution EBI, invariance tolerance).